CTSpinoPelvic1K: a fused spine and pelvis CT segmentation dataset built for lumbosacral transitional anatomy

Gregory Schwing
Department of Surgery, Detroit Medical Center / Wayne State University, Detroit, Michigan, USA

Ashley Schehr, Annika Tekumulla, Margret Khoushi, Ryan Christian, Dane Hubers, Faris Mahjoub, Hassan Saad, Mia Sooch, Sathya Siddapureddy, Michael McLellan, and Jerick Kim
Department of Surgery, Detroit Medical Center / Wayne State University, Detroit, Michigan, USA

Miraziz Ismoilov and Nizar Alnabahneh
Department of Radiology, Detroit Medical Center / Wayne State University, Detroit, Michigan, USA

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Purpose: Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present. Sacralisation and lumbarisation are different developmental events and leave different traces, but those traces are not decisive in every case, and the count that would arbitrate is taken downward from C2 — the one landmark present in every spine and never duplicated[18]. A field of view that begins in the thorax contains no such landmark, so the name rests on morphology alone exactly where morphology is least certain. Public collections either omit the pelvis (CTSpine1K[5], VerSe[3]) or omit the classes a transitional vertebra requires: TotalSegmentator[19], the largest whole-body collection, provides no sixth lumbar vertebra and no class for a rib arising from a lumbar level, and VerSe excludes vertebrae fused to the sacrum by design[4]. CTSpinoPelvic1K provides 802 CT records carrying radiologist-sourced spine *and* pelvic annotation on one coordinate frame, with per-level ribs and femora. Four classes exist so that a variant need not be forced into an ordinary level: a sixth lumbar vertebra, a thirteenth thoracic vertebra, the first sacral segment held separate from the sacrum, and left and right lumbar ribs. The primary use is the lumbosacral interface — sacral morphology measured against the lumbar spine, for pelvic fixation and lumbar surgical planning — and the transitional labelling is what makes that interface trustworthy, since a trajectory cannot be planned across a junction whose segments cannot be named. Because every record follows one screening protocol, between-scan variation reflects anatomy rather than acquisition, which suits the collection to learning morphology directly.

Acquisition and Validation Methods: Records were built by patient-level crosswalk of three public collections — TCIA CT COLONOGRAPHY imaging, CTSpine1K[5] vertebral labels, and CTPelvic1K[6] pelvic labels — with labels placed on the acquisition carrying the highest bone coverage and unified under a VerSe-native scheme. Validation is threefold: per-case geometric invariants (802/802 pass); rib–vertebra incidence across 5,749 evaluable ribs (2 residual offsets, 0.035%); and agreement of derived spinopelvic measures with published reference values.

Data Format and Usage Notes: Gzipped NIfTI image/label pairs sharing an identical affine, canonicalised to PIR, with frozen patient-grouped LSTV-stratified five-fold cross-validation splits and a reference loader. Archived at 10.5281/zenodo.22139643.

Potential Applications: Segmentation and level-numbering benchmarks, anatomical variant studies, surgical planning, and opportunistic screening research. *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; soft-tissue and hardware identifiers are declared but unpopulated; the splits are cross-validation folds and include no held-out test set; the rib layer is a reviewed pseudolabel whose review was triaged by an automated rule rather than exhaustive; and the Castellvi grades are one reader's, with confirmation by a board-certified neuroradiologist in progress.

I. PURPOSE

Labelling a lumbar vertebra requires counting, and counting requires two fixed ends. The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, but *enumeration anomalies* — eleven or thirteen thoracic, four or six lumbar — are common, and the transitional phenotypes that produce them sit at the two ends of the lumbar column: the lumbosacral transitional vertebra (LSTV) below and the thoracolumbar transitional vertebra (TLTV) above. In the presence of an LSTV the lower end is not where it is assumed to be. The two phenotypes are distinct developmental events — a lumbar segment assimilating to the sacrum, or a sacral segment separating from it — and they leave different traces: a sacralised L5 tends toward an anteriorly wedged body, a reduced disc beneath it and hypoplastic or absent facet joints, while a lumbarised S1 tends toward a squared body in the sagittal plane, well formed lumbar-type facets and a full-height disc[18]. What counting settles is the *name*. The difficulty is that these traces are not decisive in every case, so the name and the morphology can disagree, and the count that would arbitrate is taken downward from C2 — unavailable in any spine-limited acquisition. LSTV is common, reported between 4% and 30% depending on definition, and wrong-level surgery is its most serious consequence.

Existing public collections do not make this tractable, and not because they are small (Fig. 2): the spine collections omit the pelvis, the pelvic ones carry no vertebral count, and the whole-body schemes have neither an L6 class nor a class for a rib borne by a lumbar vertebra, so a six-lumbar spine cannot be written down in them at any segmentation quality. VerSe[3] is the instructive case, because it enriched for anatomical variants deliberately and does grade transitional vertebrae by Castellvi — then uses the grade to decide what to leave out. Its dataset descriptor[4] states: "Owing to the Castellvi classification we did not segment vertebrae that showed partial fusion with the sacrum (Castellvi grades III and IV)." Nor is the sacrum segmented. Those fused grades are 25 of the 33 graded records here.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra, and a rib borne by a lumbar vertebra. A scheme without those classes must record such anatomy as something it is not.

I.A. Vertebra labelling when the scan cannot settle the count

The limitation, stated plainly. Vertebra labelling is conventionally established by counting down from C2. No scan in this corpus contains C2, and none contains the whole cervical spine, so that count cannot be performed. This is not a shortcoming of the annotation: it is a property of abdominal imaging, and it means a variant at the thoracolumbar junction cannot be *resolved by counting* here at all. A thirteenth thoracic vertebra, a lumbar rib, and a *stump rib* — a hypoplastic twelfth rib, one of the cardinal indicators of a thoracolumbar transitional vertebra — produce overlapping appearances, and which label is correct depends on a count the field of view does not support.

What the release does instead. No vertebral label is asserted where it cannot be supported. Every quantity is expressed against the two landmarks present in essentially every abdominal scan — the sacrum below and the lowest rib-bearing vertebra above — so the description is positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process by its span and its distance from the ala. None of these changes according to whether a reader would have said L5 or S1, so cases that disagree about the name remain comparable. Where the junction is in view the count is determined and the conventional name follows, and the two descriptions agree.

And the morphology may be local rather than global. Counting is not the only route to an identity. A thoracic vertebra differs in shape from a lumbar one — in the span of its transverse processes relative to its body, in the proportions of its endplates and canal — and if that difference holds at the boundary itself, the phenotype is decidable from the vertebra in front of the reader without any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with each patient's own size divided out and cross-validation grouped by case, gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is deliberately removed, because a classifier given raw millimetres separates thoracic from lumbar immediately and has learned nothing that helps at the junction, where the two are nearly the same size.

That result is reported as a property of the released measurements, not as a method: it says the information needed to settle these variants is present locally, which is what makes the corpus usable for models that must assign a level without being able to count.

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

Both CTSpine1K[5] and CTPelvic1K[6] draw part of their imaging from the TCIA CT colonography collection[7,8], and the vertebral and pelvic annotations in both were produced under board-certified radiologist supervision. Neither released the mapping from an annotation to the specific CT series it was drawn on.

That mapping is not a bookkeeping detail. Every patient in CT COLONOGRAPHY was scanned *twice* — prone and supine — and each acquisition was commonly reconstructed with more than one kernel, so a single patient identifier resolves to several volumes that differ in posture, in position, in field of view and in noise. An annotation carries a patient identifier and nothing finer. Placing it on the wrong volume of the same patient produces a mask that is correctly named and, taken by itself, looks like competent work on that patient's spine — and does not sit on the bone in the volume it has been attached to.

It is not merely displaced. The spine is mildly deformable, and the two acquisitions are not two views of one posture: turning a patient from supine to prone alters lumbar lordosis and the alignment of each segment against the next, so a vertebra moves by a different

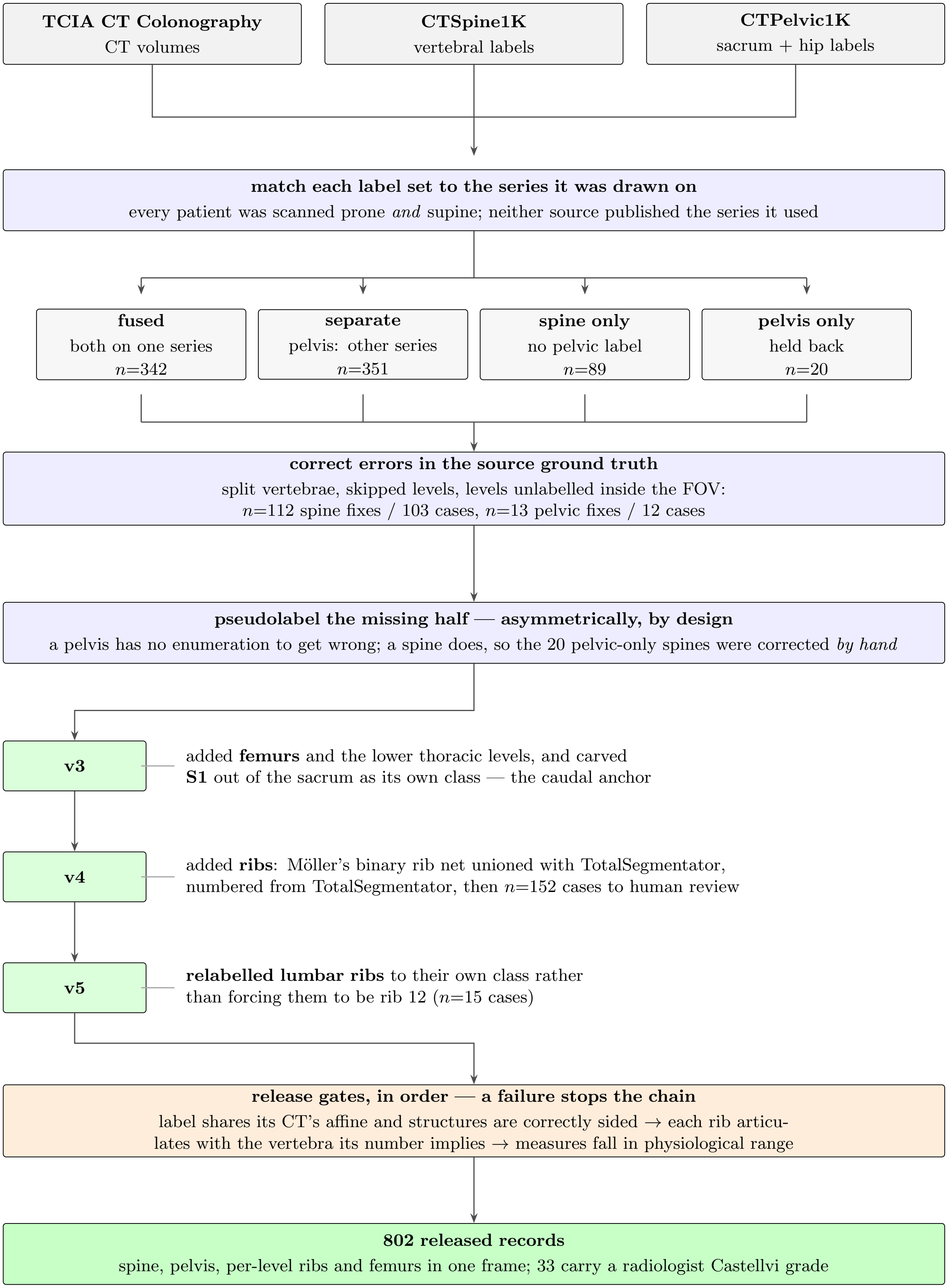

Fig. 1 How the dataset was built. Three public collections are combined patient by patient. Neither annotation source published the CT series its labels were drawn on, so each label set is matched to the series whose bone agrees with it. In 351 of those patients the two collections had drawn their outlines on *different scans of the same person*. Numbers marked n are scans, or corrections, at that step.

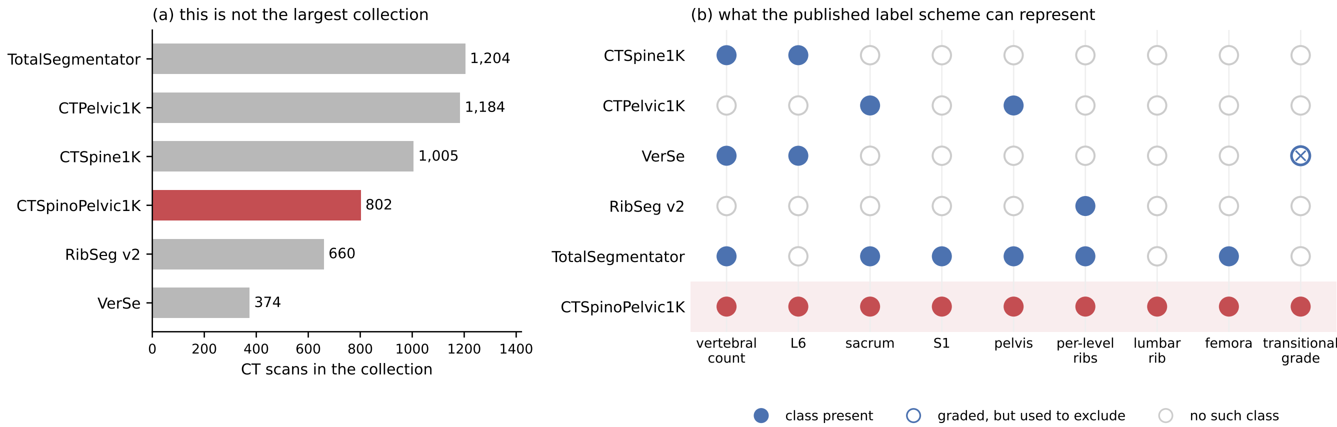

Fig. 2 Public CT collections at the lumbosacral junction. (a) Size, stated first because it is the claim this dataset does *not* make: three of the five comparators carry more scans, and RibSeg v2 labels more ribs than are labelled here. (b) What each published label scheme can represent, which is a property of its class list rather than of its segmentation quality — a scheme with no L6 class cannot record a six-lumbar spine, and a scheme that numbers ribs 1–12 has nowhere to put a thirteenth except on top of the twelfth. VerSe is marked as graded-but-excluding because it does apply the Castellvi classification and uses it to decide what to leave out: vertebrae partially fused to the sacrum, grades III and IV, are not segmented, and the sacrum is not segmented at all[4]. Those grades are 25 of the 33 graded records here. CTPelvic1K's lumbar spine is a single undifferentiated class, so no vertebral count is recoverable from it; TotalSegmentator's labels are produced by iterative pseudolabelling with manual refinement rather than drawn under radiologist supervision.

amount, and in a different direction, from the vertebra above it. Outlines drawn on one series are therefore the wrong *shape* for the other and not just in the wrong place, and no rigid realignment of the mask as a whole recovers them. That is what makes the error both hard to see and impossible to repair afterwards: in isolation the mask looks right, it reveals itself only when overlaid on the image, and by then the only remedy is to assign it to the series it was actually drawn on.

The crosswalk therefore proceeds in two stages. Each annotation is first resolved to a patient, and then, within that patient's volumes only, to the series whose bone *agrees* with it: candidate volumes are thresholded at bone attenuation and scored by overlap with the annotation mask, and the annotation is assigned to the series that maximises it. Bone is used rather than image similarity because bone is what the annotation describes; two acquisitions of the same abdomen resemble each other closely everywhere except in where the skeleton sits.

II.B. Fused and split records

The two annotation sets were placed independently, so they do not always land on the same series, and the four resulting record types are distinguished because the distinction is the crosswalk's main finding rather than bookkeeping. Of 802 records, 342 (42.6%) carry spine and pelvic annotations on the same volume and are exported as *fused*. A further 351 (43.8%) are *separate*: a pelvic annotation exists for the patient, but it was placed on the other acquisition of the prone/supine pair, so it cannot be combined with the spine annotation without resampling across a change of posture. Only 89 (11.1%) are *spine_only*, meaning no pelvic annotation exists for that patient at all, and 20 (2.5%) are *pelvic_native*, carrying the pelvic annotation alone.

The 351 separate records are the direct measurement of the problem described above: in nearly half of all patients the two public collections annotated *different volumes of the same person*, and neither collection recorded which. They are retained rather than discarded, and held out of the postural analyses, because a prone and a supine acquisition of one patient differ in spinal alignment and are a resource for a mobility study rather than a defect.

II.C. Densification, and why it is asymmetric

A corpus in which half the records lack a pelvis is not usable as a spinopelvic dataset, so the missing structures were pseudolabelled. The two directions are not equivalent and were not treated as such.

Pelvis onto spine_only records. The sacrum and the innominate bones are single objects with no enumeration: there is no count to get wrong, no naming convention to choose, and no analogue of a transitional vertebra. A pelvic segmenter can therefore be trusted in a way a vertebral one cannot, and its quality is reported as held-out performance on the *pelvic_native* records, which were withheld from training for exactly this purpose.

Spine onto pelvic_native records. The reverse direction carries the whole problem this dataset exists to address. A pseudolabelled spine must commit to a vertebral count, and on a transitional vertebra it will commit to the commoner reading. Those 20 records were therefore not densified automatically: each pseudolabelled spine was reviewed and corrected by hand. Twenty cases is a tractable amount of manual work; four hundred would not have been, which is the practical reason the asymmetry runs the way it does.

II.D. Ribs

No public rib annotation covers this cohort, and the two available automatic sources fail in complementary ways.

Möller's binary rib network[20] is a detection model of high reported accuracy, but it segments ribs that lie *entirely within* the field of view; a rib clipped by the edge of an abdominal acquisition is liable to be missed altogether. Since this is a colonography cohort, the lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

TotalSegmentator[19] does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone: the most medial portion of the rib — roughly the last tenth to sixth approaching the costovertebral joint — is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the incidence test that assigns a rib to its level has nothing to test.

The two were therefore unioned: TotalSegmentator supplies numbering and the medial-clipped cases, Möller supplies detection where its output is more complete, and the union is carried forward with TotalSegmentator's per-level identities.

The result is a pseudolabel, and the review it received was triaged rather than exhaustive — which is worth stating plainly, because a rib layer described only as "human-corrected" invites the reader to assume every bone was looked at. A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with; a single bone with two numbers is a numbering error by construction, whatever the anatomy. That rule selected 152 records. 148 were reviewed and corrected by a trained reviewer; 4 were not, and are identified in the release. Reviewers were medical students working through OpenSpineConsortium[1], an open-source framework for engaging medical students in computational spine imaging research: annotators are trained against a written labelling protocol, work case-by-case in a shared review tool, and every correction is recorded against the annotator who made it, so the review is auditable rather than anonymous. The remaining records passed the rule and were not individually inspected, so the guarantee the rib layer carries is that it is free of the failure modes the rule detects, and no stronger. The residual rib–vertebra offsets reported in Sec. II.F are the independent check on what the rule missed.

II.E. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45 left, 46–57 right), soft tissue (58–73), lumbar ribs (74–75) and surgical hardware (76–82).

Two anchors make the count decidable. Rostrally, **T12** is the last rib-bearing vertebra and sits directly above ground-truth L1. Caudally, **S1** is carved as a distinct class, giving the bottom bracket and the sacral endplate landmark required for sacral slope and pelvic incidence. With both brackets present, L5 and L6 are distinguishable where a spine-limited view alone would not be.

Neither anchor is novel on its own. TotalSegmentator[19] also carries a separate S1 class alongside the sacrum and labels ribs per level, and VerSe[3] carries L6 — this release uses the VerSe identifiers, so its L6 *is* VerSe's, and is interoperable with it and with CTSpine1K rather than local to this work. One class below has no counterpart in a published scheme; the other is included because the whole-body schemes omit it.

A rib on a lumbar body receives its own class rather than being forced to be a twelfth rib, and this is the class with no published counterpart. A lumbar rib and a stump rib are the same object under two counts; assigning it a number would commit the label to one reading of an enumeration anomaly. Its absence is decisive elsewhere: a scheme that numbers every rib 1–12 has nowhere to put a thirteenth, so the annotator must either call it rib 12 — which asserts the vertebra beneath it is thoracic, the very question at issue — or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe's T13 is a vertebra rather than a rib. 16 released records carry a lumbar rib and 18 carry an L6.

L6 is a class, not an exception. It is inherited rather than introduced, and the reason it is retained is that a scheme without one cannot record a six-lumbar spine at all, and must renumber the column or drop a level to fit. The whole-body collections do exactly that: TotalSegmentator stops at L5.

Figure 3 shows why the interval is the primitive and the label is derived. Among the records with four rib-free vertebrae, some reach that count through a lumbar rib and some without one; the two are indistinguishable by the count alone and are distinguished here only because rib presence is itself labelled.

Sixteen records carry a lumbar rib in the released volumes: thirteen bilateral and three unilateral, all three on the right. That laterality is reported because it is what the release contains and not as a finding — sixteen records cannot support a statement about side preference, and none is made.

Eighteen records carry an L6. Fourteen are labelled lumbarisation, two sacralisation, one semi-sacralisation and one normal, which is worth stating because it shows the two annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and still be read as a sacralisation, or as unremarkable, depending on where the reader started the count.

Both counts are derived from the released label vol-

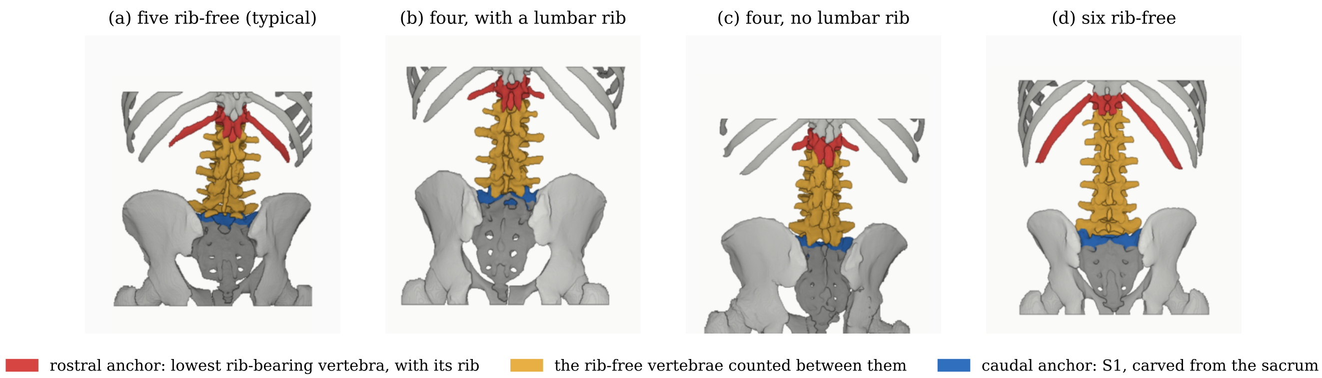

FIG. 3 The two anchors, and the interval counted between them, rendered from the released labels in a posterior view. The rostral anchor is the lowest rib-bearing vertebra together with the rib that makes it one; the caudal anchor is S1, carved from the sacrum. Both are found from the labels rather than assumed, so in (b) the rostral anchor is a lumbar vertebra bearing a rib rather than T12 — colouring by role rather than by identifier is what keeps the panels comparable, since colouring T12 specifically would assert the count the figure is deriving. Panels (b) and (c) carry the same interval of four reached from opposite ends, and are indistinguishable by the interval alone. All four panels are resampled to one millimetre scale and aligned at the sacral base, so the interval lengths compare directly rather than reflecting voxel size.

umes, and so are the manifest fields that report them. That is a correction rather than a design note: through v5 the `has_l6` flag was true for exactly one record, which contains no L6, and false for all eighteen that do, while `has_lumbar_rib` was false everywhere and `n_lumbar_labels` read zero in 799 of 802 records. A reader selecting the six-lumbar cases on those fields would have received one record that is not one, and a reader selecting lumbar ribs an empty set, with nothing to indicate either. The fields are rebuilt for this release by counting identifiers 25 and 74–75 in each volume, and `lumbar_rib_side` is added alongside them.

II.F. Validation

Validation is layered, and the ordering matters: measurements computed on a corpus that has not established internal consistency do not look broken, they look finished.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for left- and right-sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. This check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed one label away from its image — a failure invisible to every downstream step.

The test compares *each sided pair separately*, and neither alternative finds what it finds. Ribs alone passed all 802 while four records carried `left_hip` on the patient's right. Pooling the pairs — the obvious repair — recovered three of the four and missed one, because hips and femora outweigh the ribs by voxel count and a large correctly-sided structure masks a smaller transposed one. All four were corrected and the release re-verified. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs present, 5,788 are not evaluable at all — the vertebra their number implies lies outside the field of view and nothing else is within the anchor distance — and a further 11 have no vertebra in reach. That leaves 5,749 with an anchor to check, of which two remain offset (0.035%). The denominator is stated because quoting two against 11,548 halves the rate by counting ribs the check never examined, and because more than half of the ribs in a screening abdominal CT being unevaluable is itself a fact about the corpus worth reporting. Both offsets are field-of-view truncations on individually characterised cases — one at +1 level and one at −1 — and are reported rather than suppressed: a threshold tuned until the count reaches zero yields a cleaner number and a less informative one.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra — not because no lumbar ribs exist, since 15 records carry one, but because such a rib takes class 74 or 75 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Across versions. Running the same checks against every release makes the version history a measurement rather than a list of dates. Ribs arrived with fragmentation — 1146 labels in more than one piece against 366 before ribs existed — and this release carries 377 while still carrying the ribs. A fragmented label is scan truncation, which is legitimate, or speckle, which is not.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*[2] (Table III). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

II.G. Castellvi typing, and the reading it still requires

Every record whose vertebral count departs from five carries a Castellvi grade[14] in the released manifest: 33 cases, typed I–IV with the a/b unilateral–bilateral qualifier (Table II). This is released as an annotation layer because the grade and the count are *different axes* and the dataset is built to show that they are: grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A corpus that recorded only the count would describe those seven as unremarkable.

The grading is a single reader's, with five cases independently read a second time. Three of those five agreed exactly. Of the two that did not, one is Ib against IIb — adjacent subtypes, the ordinary kind of disagreement — and the other is IIIb against IIb, which crosses the boundary between bony fusion and articulation without it. That is the distinction the classification is built on and the one carrying clinical weight[18], so it is reported rather than folded into a single agreement rate. Five second reads establish that the grading was checked; they are far too few to support an inter-rater statistic, and none is quoted.

The typing is therefore submitted for confirmation by a board-certified neuroradiologist, blinded to the existing grades and to the vertebral counts, reading all 33 cases. Where the confirming read and the original agree the grade stands; where they disagree the case is adjudicated in a joint reading and the adjudicated grade released, with both original reads retained in the manifest so that no disagreement is silently resolved. Full agreement statistics will be reported for the complete set of 33 rather than for the current subset of five. *This confirmation is in progress and the release is not final until it completes*; the grades described here are the single-reader grades, and the manifest records which reads each grade rests on.

II.H. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here for a specific reason: **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**. The transitional-anatomy analysis measures the gap between the lowest lumbar vertebra and the sacrum, and a cage-bridged interspace reads as "no gap" exactly as a congenitally fused transitional vertebra does. An unrecorded instrumented case can therefore contribute a false positive to the central result, and until this release nothing in the label scheme or the manifest said which cases they were: identifiers 76–79 were declared in every prior version and populated in none.

II.H.0.a. Detection over-calls by a factor of seven. A threshold at 1800 HU inside a shell around the labelled skeleton flags 84 of the 802 records. Raising it to 2500 HU — the lower of the two values validated in the metal-segmentation literature — leaves 52 with a component above a 40-voxel floor. A radiologist read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Instrumentation prevalence in this cohort is 1.4%, not the 10.5% an unreviewed threshold reports. The threshold at 3000 HU is not the answer either: these series clip at 3071 HU, so it measures the saturation plateau rather than the implant.

II.H.0.b. Saturation does not separate implant from artefact. This is the obvious test and it fails. Both groups reach the scanner ceiling, and 4 of the rejected proposals *exceed* it, peaking at 11,798 HU — values above the ceiling being the signature of reconstruction overshoot around something dense rather than of denser metal. What does separate them is position and size together: a real surgical site (a hip joint, the sacroiliac joint, the spine) *and* volume, with every confirmed implant at 2,586 mm^3 or above and every rejection at 1,768 mm^3 or below (Fig. 4).

II.H.0.c. The subtype block had to be extended. Identifiers 76–79 were written for spinal instrumentation — generic, cage, screw-and-rod, plate — and cannot name what this cohort holds. Three classes were added: **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis**. The last two are not decoration. Osteosynthesis holds parts of the same bone together where arthroplasty replaces a joint, a dichotomy the femoral-neck literature is written in, and the shape rules would otherwise have called a femoral stem a rod, because a stem is long and thin and that is all "linear" tests. The confirmed cases comprise 8 arthroplasties, one osteosynthesis (three parallel cannulated screws in an inverted-triangle configuration at the femoral neck), one sacroiliac screw fixation and one pair of threaded cylindrical interbody cages.

II.H.0.d. Metal outranks bone, and that is a subtraction. In every confirmed case the implant was already inside a bone label, a segmenter handed a bright object against a cortical surface having taken it for bone. Naming the hardware therefore reclaims voxels rather than adding them: 1,538,852 voxels across the eleven records. The consequence reaches past bookkeeping. Pelvic incidence and pelvic tilt are measured from the femoral head centre, and in 8 of these cases that head is an implant — any spinopelvic parameter previously computed from them was measured on metal.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canonicalised to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the downstream tooling expects.

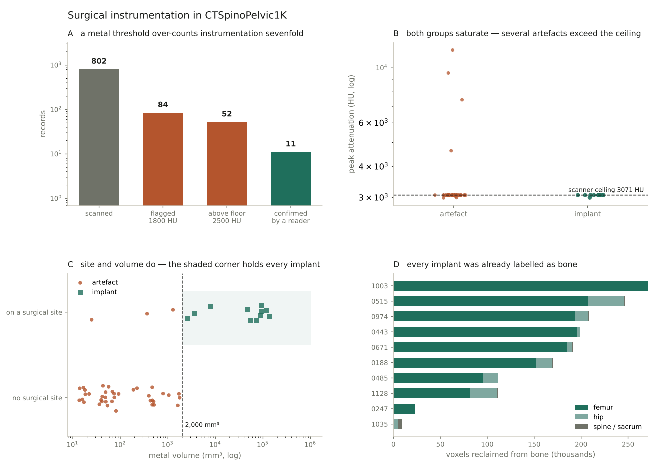

Fig. 4 Surgical instrumentation. (a) An unreviewed metal threshold over-counts instrumentation sevenfold. (b) Saturation cannot sort implant from artefact: both reach the scanner ceiling and several artefacts exceed it. (c) Site and volume can. (d) Every confirmed implant was already labelled as bone, so naming it reclaims voxels from the structure that held them.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as `splits_5fold.json`. The five validation folds partition all 802 records — they are disjoint and their union is the cohort — so the release provides cross-validation and *not* a held-out test set. Anyone reporting a single held-out number should carve one out and say which records it holds, because the folds shipped here will not do that job.

Stratification is on the transitional subtype rather than on a binary LSTV flag, and it enforces a minimum of each rare subtype in every validation fold: each fold's validation set carries three sacralisation and three lumbarisation records, so no fold is evaluated on a cohort missing the class the dataset exists for. With 15 lumbarisation and 15 sacralisation records across 802, that is the practical limit of what stratification can guarantee, and it is the reason a fold-level metric on the rare classes has an error bar far wider than its own decimal places.

The archive of record is Zenodo (10.5281/zenodo.22139643). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM's *other* value for patient sex, which is a recorded value and not a missing one, and 53 carry no sex field at all. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from measures stratified by sex — a limitation of those measures rather than a property of the cohort, and stated here because folding eleven people into "unrecorded" would misdescribe what the source says. The lower age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer to a reference range than a surgical series would.

Count-free measures (Fig. 5). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, with modes near 0.33 and 0.69.

Level gradients and spinopelvic measures (Fig. 6). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

Opportunistic measures (Fig. 7). Every scan carries a vertebral bone density measurement at no additional dose. L1 trabecular attenuation is reported at the published standard site.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*: spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, which makes sacral morphology measurable against the lumbar column rather than in isolation — the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads rather than from landmarks placed on a radiograph, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it. A corpus assembled purely for enumeration anomalies would want whole-spine coverage this cohort lacks; what these records support is planning *across* a junction, and no trajectory can be planned across a junction whose segments cannot be named. Level-numbering benchmarks, variant studies and opportunistic screening (Fig. 7) follow from the same annotation.

V.A. Three uses the released measurements support

Naming a level from its own geometry. Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures and 4.5–12.8 per 10,000 lumbar discectomies annually, and half of surveyed neurosurgeons report having operated at the wrong level at least once; the cause most often cited is the transitional anatomy this cohort was assembled around.[11] The failure is structural rather than careless, since the count is ambiguous exactly where the variant sits. Whether a level can instead be named from the local morphology of the vertebra in front of the reader is a question this release is shaped to support: per-level body height, canal width, endplate width, transverse-process span and wedge ratio are released for

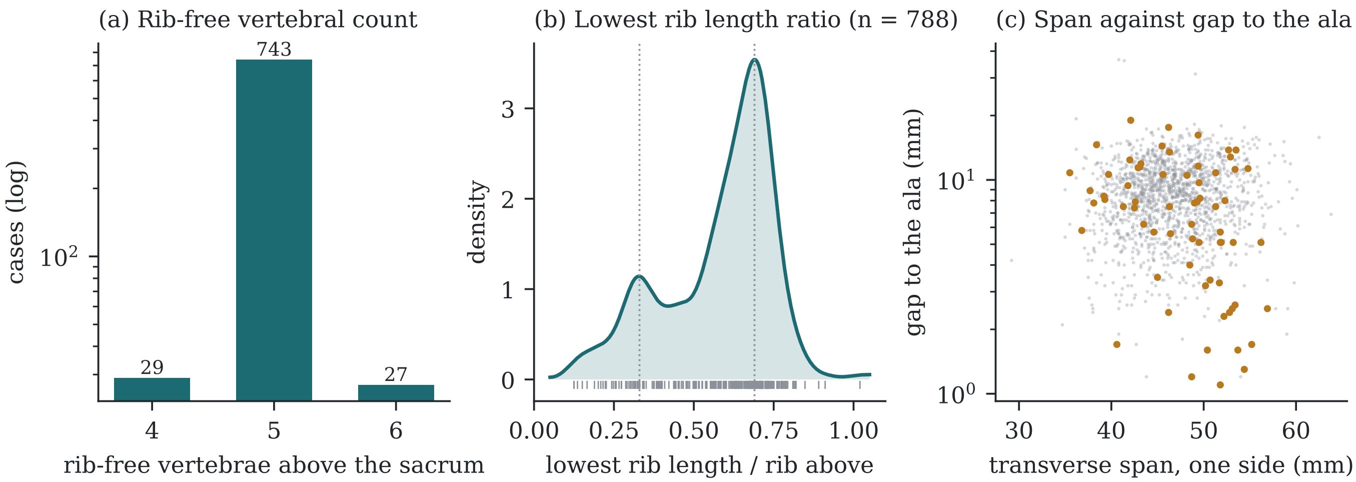

Fig. 5 Count-free measures, valid without knowing which vertebra is which. (a) Vertebrae between the lowest rib and the sacrum: an interval, not a level assignment. (b) Lowest rib length as a fraction of the rib above, showing two populations near 0.33 and 0.69 rather than one distribution with a tail. (c) Lowest lumbar transverse span against its gap to the sacral ala; cases carrying a source transitional label are marked.

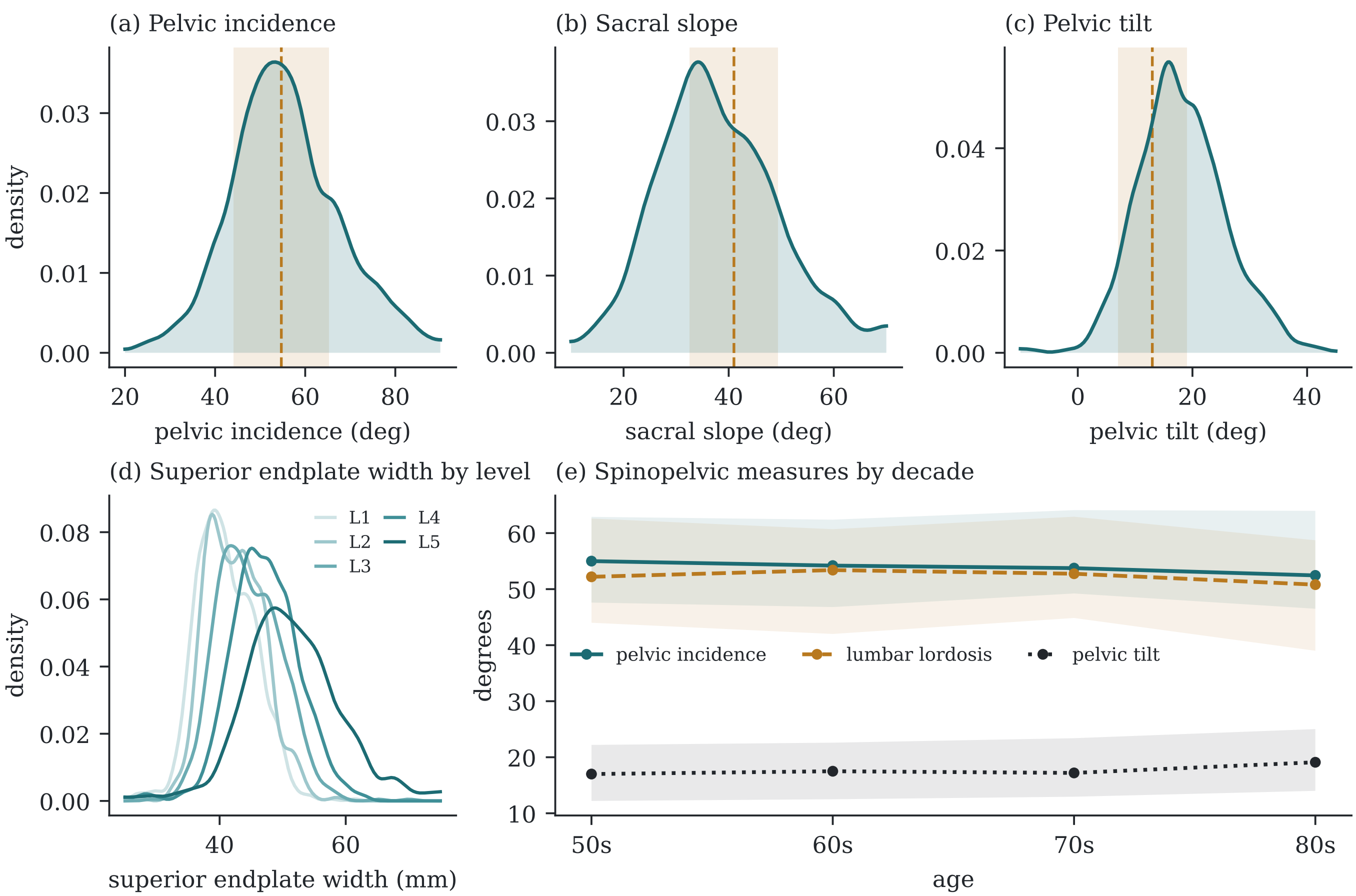

Fig. 6 (a–c) Spinopelvic measures against published reference bands (mean ± SD). (d) Superior endplate width by level. (e) Median and interquartile range by decade; pelvic incidence is flat by construction of the anatomy, which makes it a negative control for the measures beside it.

T11–L5 across all 802 records, the labels were assigned against the twelfth rib rather than by enumeration — so a model trained here is not learning the convention that fails — and the transitional records supply the cases on which a count-based method and a geometric one should disagree.

Reference morphometry at cohort scale. Reference values for vertebral and canal dimensions derive substan-

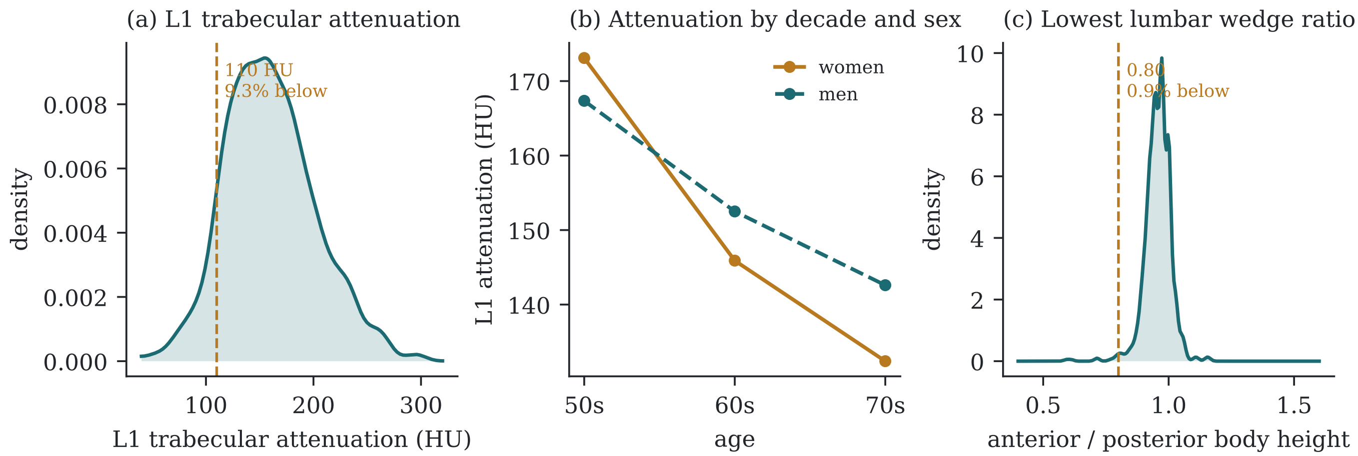

Fig. 7 Opportunistic measures from the same images. (a) L1 trabecular attenuation, median 157 HU, against the 110 HU osteoporosis-specific threshold. (b) Median attenuation by decade and sex. (c) Lowest lumbar wedge ratio with Genant[10]'s 0.80 threshold for mild wedge deformity.

tially from cadaveric series and small imaging cohorts.[12] These measurements are computed identically across 802 records from segmentations rather than from landmarks placed by hand. Larger n improves the precision of a distribution and does not fix the population it was drawn from: this is a screening cohort aged 50 and over, imaged supine, so it updates the sample size behind a reference range rather than making the range representative.

Patient-specific models. Planning is moving from static radiographic targets toward patient-specific biomechanical models, on the argument that aligning a patient to a population norm is the wrong objective when their own geometry is measurable.[13] Such a model needs the spine, pelvis and femoral heads in one frame, which is this release's construction, and pelvic incidence, tilt and sacral slope derive here from the segmented sacrum and femoral heads rather than from radiographic landmarks. In 351 patients the annotation sits on two acquisitions in different positions, giving a within-patient change in alignment against which a predicted postural response can be checked — the difference between a model that is built and one that is validated — and the eleven instrumented records carry hardware as its own structure, so a post-operative state is representable. These remain supine screening scans of an older cohort: they support building and checking such models, not setting the alignment targets a correction should aim for.

Its limitations are part of it, and they are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

Declared but empty, and confirmed so. The soft-tissue identifiers (58–73) are declared in the scheme and occur in no record — checked by counting identifiers in all 802 released volumes rather than by assertion. They are retained so that a downstream annotation can use them without renumbering, and a user should treat their absence as absence of annotation rather than absence of the structure. The hardware identifiers were in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented records described in Sec. II.H.

Strength of the labels varies by structure, and the release does not average over that. Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong. Pelvic labels on records that lacked one are pseudolabelled. The rib layer is a pseudolabel whose human review was triaged by an automated rule rather than exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional labels derive from two independent sources and are not uniformly adjudicated — which is precisely why the measures reported here are count-free — and the Castellvi grades are a single reader's pending the confirmation described in Sec. II.F.

Structures are not universally present. The same census shows the release is not uniformly complete: one record carries no sacrum, hips or femora, and two carry no S1. Two lack T12 and nine lack an L5 identifier, in every case because the structure lies outside the field of view or, for L5, because the lowest lumbar segment is labelled L6 or incorporated into the sacrum. A user filtering on the presence of the caudal anchor should filter explicitly rather than assume it.

Splits. The shipped folds are cross-validation folds covering the whole cohort. There is no held-out test set, and with 15 lumbarisation and 15 sacralisation records a per-fold metric on the rare classes is estimated from three cases.

DATA AVAILABILITY

The dataset is permanently archived at https://doi.org/10.5281/zenodo.22139643. Code is at https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K, tagged at the release commit.

CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

ETHICS

This work uses publicly available, de-identified imaging from The Cancer Imaging Archive[7,8] and annotations derived from it. The investigators obtain no identifiable private information and have no interaction with the individuals imaged, so the activity does not meet the definition of human participant research under 45 CFR 46.102; a Wayne State University Human Participant Research determination to that effect has been completed and is retained by the investigators.

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TABLE I Release validation. Gates run in order and a failure stops the chain.

Check	Result
Geometric invariants (all cases)	802/802 pass
label shape, affine, spacing match CT	—
no identifier outside the scheme	—
left/right structures correctly sided	—
Ribs present	11,548
not evaluable (expected vertebra outside the FOV)	5,788
no vertebra within the anchor distance	11
evaluable	**5,749**
matching the expected vertebra	5,747
residual offset	2 (0.035% of evaluable)
misnumbered cases	0
Spinopelvic medians within published range	6/6

TABLE II Radiologist Castellvi grade against the rib-free vertebral count, for the 33 graded records. The two are different axes and the table shows how far apart they run: grade IIIb occurs at counts of four, five *and* six, and seven graded records carry the normal count of five, where nothing about the count would draw attention.

		rib-free vertebrae		
Castellvi grade	records	4	5	6
Ib	2	0	0	2
IIa	2	0	1	1
IIb	4	0	0	4
IIIa	3	0	1	2
IIIb	18	7	4	7
IV	4	1	1	2
all	33	8	7	18

TABLE III Derived measures against a published asymptomatic reference ($n = 260$, standing)[2]. Sacral slope and pelvic tilt are postural where pelvic incidence is not, and this cohort is supine. The two postural measures depart from the standing reference in opposite directions — sacral slope 4.1° lower, pelvic tilt 4.3° higher — which is not two discrepancies but one, because $PI = PT + SS$ holds by construction: with pelvic incidence fixed at its published value, any fall in sacral slope must appear as an equal rise in pelvic tilt. That the two offsets match in magnitude is evidence the geometry is internally consistent rather than evidence of error.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI − LL mismatch	2.6°	centred near 0